Statistical Thinking for Data Science

Week 5: Describing relationships

Two variables, and the third one you forgot

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Week 5: Describing relationships

Learning Objectives

By the end of this week you should be able to:

- read a two-way table, and compute the conditional percentages that answer the question actually asked;
- compute and distinguish a risk difference, a risk ratio and an odds ratio;
- recognise Simpson's paradox and explain what produces it;
- compute a covariance and a correlation by hand, and explain the standardisation;
- list four situations in which a correlation is misleading, and say which way;
- compare groups with a difference in means and a standardised effect size;
- define a confounder, draw the situation, and check for it by stratifying.

5.1 Four kinds of pair

Describing a relationship starts with asking what kinds of variable you have, because that decides the tool.

Pair	Picture	Summary
Categorical, categorical	Two-way table, stacked bars	Conditional percentages, risk difference, risk ratio, odds ratio
Numeric, numeric	Scatterplot	Covariance, correlation, later a regression line
Categorical, numeric	Box plots side by side	Difference in means or medians, standardised effect size
Anything against time	Line plot	Trend, seasonality, change points

5.2 Two categorical variables

Example 1: tutorials and passing

Among 300 students, 120 attended the optional tutorials and 180 did not.

	Passed	Failed	Total
Attended tutorials	96	24	120
Did not attend	108	72	180
Total	204	96	300

Solution

First the conditional percentages, which is what the question is about: $96/120 = 80\%$ of tutorial-goers passed, against $108/180 = 60\%$ of the others. Three summaries of that gap:

$$\text{risk difference} = 0.80 - 0.60 = 0.20, \qquad \text{risk ratio} = \frac{0.80}{0.60} = 1.33,$$

$$\text{odds ratio} = \frac{96/24}{108/72} = \frac{4.00}{1.50} = 2.67.$$

The odds ratio is much the largest, as it always is when the outcome is common, and it is the one most often quoted by people who want the number to sound big. The risk difference is the one a Head of Department can act on: twenty students in every hundred.

The warning

None of this says tutorials caused anything. Students who choose to attend tutorials differ from those who do not, in motivation, in timetable, in how much paid work they do. Section 5.6 is about exactly this.

In a two-way table, always condition in the direction of the question. "What percentage of tutorial-goers passed?" and "what percentage of those who passed went to tutorials?" are different numbers, and swapping them is the most common error in student reports.

5.3 Simpson's paradox

A relationship can hold in every subgroup and reverse when the subgroups are pooled. This is not a curiosity; it happens whenever the groups differ in both their mix and their outcome.

Example 2: the method that is better in both programmes and worse overall

Two teaching methods are trialled in two programmes.

	Method X		Method Y	
	Passes / students	Rate	Passes / students	Rate
Programme A	80/100	80%	18/20	90%
Programme B	4/20	20%	30/100	30%
Combined	84/120	70%	48/120	40%

Solution

Method Y beats method X in Programme A (90% against 80%) and in Programme B (30% against 20%). Pooled, method X appears far better: 70% against 40%.

The arithmetic is sound; the pooling is the problem. Programme A is an easier course (high pass rates under both methods) and most of method X's students sat it, while most of method Y's students sat the harder Programme B. The pooled comparison is mostly a comparison of programmes wearing the label of a comparison of methods.

What to do

Report the subgroup results, or a weighted average using a common set of weights. Never report the pooled figure alone when the mix differs. The general question to ask is the one from Week 2: is the comparison being made within a group, or across groups that differ?

5.4 Two numeric variables

5.4.1 Look first

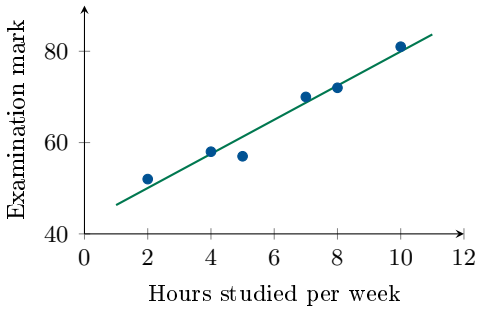


Figure 5.1: Six students. The relationship is close to linear and strongly positive, which is what makes a correlation a reasonable summary here.

5.4.2 Covariance

Definition 5.1 (Sample covariance)**.**

$$\operatorname{Cov}(x, y) = \frac{1}{n-1} \sum_{i=1}^{n} (x_i - \bar{x})(y_i - \bar{y}).$$

Each term is positive when both values sit on the same side of their means, negative otherwise, so the sum measures whether the two move together. Its weakness is units: the covariance of hours with marks is in hour-marks, and changing hours to minutes multiplies it by 60 without changing anything real.

5.4.3 Correlation is standardised covariance

Definition 5.2 (Pearson correlation)**.**

$$r = \frac{\operatorname{Cov}(x, y)}{s_x s_y} = \frac{\sum_i (x_i - \bar{x})(y_i - \bar{y})}{\sqrt{\sum_i (x_i - \bar{x})^2} \sqrt{\sum_i (y_i - \bar{y})^2}}.$$

Dividing by the two standard deviations cancels the units, which forces $-1 \le r \le 1$ and makes r unchanged by any change of scale or origin.

Example 3: a correlation by hand

The six students in the figure:

Hours x	2	4	5	7	8	10
Mark y	52	58	57	70	72	81

Solution

$\bar{x} = 6$ and $\bar{y} = 65$. Building the products of deviations:

$$(-4)(-13) + (-2)(-7) + (-1)(-8) + (1)(5) + (2)(7) + (4)(16) = 52 + 14 + 8 + 5 + 14 + 64 = 157,$$

so $\operatorname{Cov}(x, y) = 157/5 = 31.4$. With $s_x = 2.898$ and $s_y = 11.063$,

$$r = \frac{31.4}{2.898 \times 11.063} = 0.979.$$

Then $r^2 = 0.959$: about 96% of the variation in marks lines up with variation in study hours *in this sample of six*, which is a very small sample and a very tidy one.

r measures the strength of a **straight-line** relationship only. It says nothing about the slope: $r = 0.9$ is consistent with a steep line or a nearly flat one. For how much y changes per unit of x, you need a regression, which is Week 11.

5.5 Four ways a correlation misleads

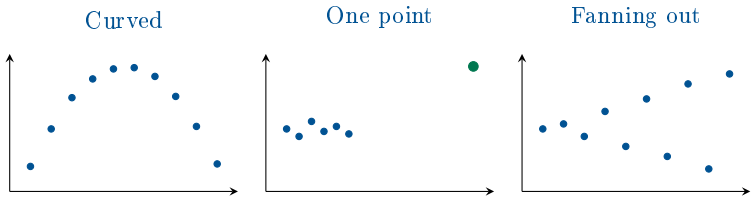


Figure 5.2: Three datasets a single correlation coefficient describes badly: a strong curved relationship with r near zero; a flat cloud plus one far point, which alone manufactures a correlation; and a fan, where the relationship's *spread* changes with x.

5.5.1 Non-linearity

A perfect curved relationship can have $r = 0$. Always plot before you trust r.

5.5.2 A single influential point

Example 4: one student changes everything

Add one more student to Example 3: 3 hours of study, mark 95. Perhaps they had seen the paper, perhaps the mark was mistyped.

Solution

The correlation collapses from $r = 0.979$ to $r = 0.305$. One observation in seven has taken a near-perfect relationship and made it look weak.

The same works in reverse: adding one extreme point to an uncorrelated cloud can produce $r = 0.8$, as the middle panel of the figure shows. Whenever you report a correlation, check what happens when the most extreme point is removed, and say so if it matters.

5.5.3 Restriction of range

Example 5: why entrance scores look useless to selective programmes

Twelve applicants have entrance scores and later final marks:

Entrance	20	28	35	41	46	52	58	63	70	76	84	92
Final	41	48	55	58	63	66	72	80	74	86	76	82

Across all twelve, $r = 0.935$, so $r^2 = 0.87$. Now suppose the programme admits only applicants scoring 58 or more, and someone studies the admitted students alone. Among those six, $r = 0.468$ and $r^2 = 0.22$.

Solution

Nothing about the applicants changed. Selecting on x cut its standard deviation from 22.5 to 12.8, and the correlation fell with it. This is why admissions tests routinely look weak when evaluated on admitted students only, and why an argument of the form "our entrance test does not predict performance" needs checking against the full applicant pool before it is believed.

5.5.4 Changing spread

If the scatter widens as x grows, a single r averages over two regimes: tight at one end, loose at the other. The correlation is not wrong, it is just a poor summary, and any prediction interval built from it will be too wide at one end and too narrow at the other.

5.6 One categorical, one numeric

Compare group summaries, and report the difference with its context.

Example 6: a difference in means, standardised

Forty students who used the practice sheets average 72 marks; forty who did not average 65. Both groups have $s = 12$.

Solution

The difference is 7 marks. To express it on a scale that travels between studies, use the pooled standard deviation and Cohen's d:

$$s_p = \sqrt{\frac{(n_1 - 1)s_1^2 + (n_2 - 1)s_2^2}{n_1 + n_2 - 2}} = 12, \qquad d = \frac{72 - 65}{12} = 0.58.$$

A d of about 0.6 is a moderate effect: the groups overlap heavily, and plenty of non-users beat plenty of users.

Say both

Report the raw difference, because 7 marks is what a student experiences, and the standardised one, because it is what lets you compare this with a study measured on another scale. Never report d alone to a practical audience: nobody has an intuition for it.

5.7 Confounding: the third variable

Definition 5.3 (Confounder)**.** A variable associated with both the supposed cause and the outcome, which can produce an association between them even if neither affects the other.

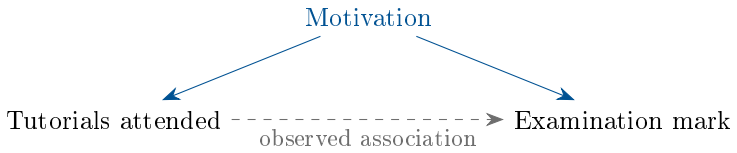


Figure 5.3: The dashed arrow is what you measured. The solid arrows are what may have produced it. No amount of extra data on the two bottom variables can tell them apart.

The practical check is to **stratify**: compute the relationship separately within levels of the suspected confounder. If the association survives within every stratum, it is not explained by that variable; if it vanishes, it was. That is exactly what Example 2 did, and what Week 2's two shifts did.

Do not control for everything. Adjusting for a variable that sits on the causal path between cause and effect removes part of the very effect you are measuring, and adjusting for a common *consequence* of both can create an association out of nothing. Choose the variables to adjust for from knowledge of the subject, never by whatever improves the fit.

5.7.1 From association to cause

When an experiment is impossible, an observational claim is strengthened by:

- the supposed cause coming first in time;
- a dose-response pattern, more exposure giving more effect;
- the association surviving adjustment for the obvious confounders;
- a mechanism that makes physical sense;
- the same result appearing in different places with different biases.

None of these is proof. Together they are how most of what we know about health, safety and reliability was established, because the experiment was unethical, impossible or unaffordable.

5.8 Reporting a relationship

> "Among 300 students, 80% of those attending tutorials passed, against 60% of those who did not: a difference of 20 percentage points (risk ratio 1.33). Attendance was not assigned, and tutorial-goers also reported more study hours, so this is an association, not an effect of the tutorials."

Three ingredients: the numbers conditioned the right way, the comparison stated in a scale people can act on, and an explicit sentence about what the design does and does not support.

5.9 Doing it on a computer

R

```
tab <- table(tutorial, result)
prop.table(tab, margin = 1)          # row percentages: condition on tutorial

cor(x, y)                            # Pearson
cor(x, y, method = "spearman")       # rank-based, resists outliers
plot(x, y); abline(lm(y ~ x))

by(mark, group, summary)             # numeric by category
```

Python

```
pd.crosstab(tutorial, result, normalize="index")   # row percentages

df[["x","y"]].corr()                 # Pearson
df[["x","y"]].corr(method="spearman")
df.plot.scatter("x", "y")

df.groupby("group")["mark"].describe()
```

5.10 A checklist for any claimed relationship

1. Plot it before summarising it.
2. Which way are the percentages conditioned?
3. Would the relationship survive inside subgroups?
4. What happens if the most extreme point is removed?
5. Was the sample selected on one of the two variables?
6. Is the relationship straight, or have I summarised a curve with one number?
7. Who was compared with whom, and could they differ in some other way?

5.11 Glossary

Conditional distribution	The distribution of one variable within a fixed level of another.
Risk difference	$p_1 - p_2$, in percentage points.
Risk ratio	p_1/p_2.
Odds ratio	$\dfrac{p_1/(1-p_1)}{p_2/(1-p_2)}$; exceeds the risk ratio when outcomes are common.
Covariance	Average product of deviations; units are the product of the two variables' units.
Correlation r	Covariance divided by both standard deviations; unit-free, between -1 and 1.
r^2	Share of variance shared by the two variables.
Cohen's d	Difference in means divided by the pooled standard deviation.
Confounder	Associated with both variables, and capable of creating the association.
Simpson's paradox	A reversal between subgroup and pooled comparisons.

5.12 Summary

Week 5 in six lines

- Choose the tool from the pair of variable types, and plot before summarising.
- Condition percentages in the direction of the question.
- A pooled comparison can reverse the subgroup comparisons; report the subgroups.
- r measures straight-line strength only, and one point can create or destroy it.
- Selecting on a variable shrinks its correlations, which is not the same as weakening the relationship.
- An association plus a plausible third variable is not an effect.

Exercises

1. From Example 1's table, compute the percentage of those who passed who had attended tutorials. Explain why it answers a different question from 80%.
2. In a sample, 45 of 60 machines serviced weekly ran without failure, against 30 of 90 serviced monthly. Compute the two rates, the risk difference, the risk ratio and the odds ratio.
3. Construct your own small example of Simpson's paradox, with two groups and whole numbers, and explain which difference in mix produces it.

4. For the data $x = 1, 3, 4, 6, 8$ and $y = 2, 5, 4, 9, 12$, compute $\bar{x}$, $\bar{y}$, the covariance and r by hand.
5. In Exercise 4, convert x from metres to centimetres and recompute the covariance and r. Which one changed, and why?
6. Sketch a scatterplot with r close to zero in which the two variables are nevertheless perfectly related.
7. A study of admitted students finds $r = 0.2$ between entrance score and final mark, and concludes the entrance test is useless. Give two reasons to doubt the conclusion.
8. Two groups have means 48 and 56 with $s = 20$ in both and $n = 50$ each. Compute Cohen's d and describe the overlap in plain words.
9. A manager observes that shifts using the new procedure have fewer incidents. Name three confounders, and say how you would check each by stratifying.
10. Explain, with an example from your field, why adjusting for a variable on the causal path is a mistake.
11. In the class wellbeing data, state two pairs of variables you could relate, and for each the right picture and summary.
12. Take any two numeric variables from a dataset you have. Plot them, compute r, remove the most extreme point, recompute, and report both in one sentence.

Answers to the odd-numbered exercises

1. $96/204 = 47\%$. This conditions on passing rather than on attending, so it is shaped by how many students chose each option; with twice as many non-attenders in the class, it can be below half even though attenders pass more often.
3. Any example where the groups differ in both mix and baseline rate will do. The check is that the subgroup rates favour one option while the pooled rates favour the other, which requires the better option to be concentrated in the harder subgroup.
5. $\bar{x} = 4.4$, $\bar{y} = 6.4$, covariance $= 10.55$, $r = 0.967$. In centimetres the covariance becomes 1055, a hundred times larger, while r stays 0.967: the standardisation removes the units.
7. First, restriction of range: among admitted students the entrance scores barely vary, which shrinks r mechanically (Example 5 fell from 0.935 to 0.468 on exactly this). Second, the admitted group has been selected partly on the test itself, so the comparison is no longer about the applicant population the test is used on.
9. For instance operator experience, equipment age and the type of work done on each shift. For each, split the data into levels of that variable and compare incident rates within levels; if the new procedure still looks better inside every level, that confounder does not explain the result.
11. For example: sleep hours against the depression subscale (box plots by sleep band, difference in medians), and programme against anxiety score (box plots by programme, difference in means with an effect size).

Before next week

Read *OpenIntro Statistics* Section 2.3 and bring the output of Exercise 12. Week 6 is about the pictures themselves: what makes a graph argue a point clearly, and the standard ways graphs mislead.